Major Project (ETMJ100) Presentation

Title of Major Project

AI-Powered Chest X-ray Multi-Label Classification and Impression Generation Using Medical Vision and Language Models

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INTRODUCTION

Chest X-ray (CXR) remains the most frequently performed diagnostic imaging examination worldwide. However, its interpretation demands highly skilled radiologists, leading to significant time delays, inter-observer variability, and increasing workload in resource-constrained healthcare systems. While traditional deep learning models have achieved high accuracy in multi-label classification of 14 common thoracic pathologies, they typically provide only binary or probabilistic labels with limited clinical explainability.

MOTIVATION

Radiologists, in real practice, rely heavily on:

- Comparison with similar historical cases
- Explicit documentation of deviations from normal anatomy
- Structured impression sections that summarize key findings

Existing automated systems fall short in delivering such contextual, human-readable, and factually grounded assistance. Therefore, there is an urgent need for an AI-powered system that goes beyond classification to offer **multi-label disease prediction, retrieval of visually and clinically similar cases, and generation of accurate, explainable radiology impressions** — all within a single integrated framework.

OBJECTIVE

- To develop an end-to-end system for **multi-label classification** of various pathologies in chest X-rays using medical vision-language embeddings.
- To design a **multimodal Retrieval-Augmented Generation (RAG)** framework that retrieves clinically similar past cases using joint image–report embeddings.
- To introduce a **novel contrastive healthy baseline guidance mechanism** that computes embedding deviation from normal CXRs to emphasize abnormalities during retrieval and generation.
- To generate **clinically accurate, hallucination-reduced impressions and findings** using a fine-tuned large language model conditioned on retrieved context and healthy deviation signals.

DATASET

•Primary Dataset: MIMIC-CXR
377k CXR images + reports from 227k studies; multi-label annotations (e.g., CheXpert classes like atelectasis, edema).
Filter healthy normals (~30% via labels) for baseline; extract findings/impressions; preprocess images (resize to 224x224).

•Secondary Dataset: IU X-Ray
7k images + reports; focused on frontal/lateral views; used for evaluation/transfer.

KEY FEATURES

ASET

- Multimodal embeddings using **BioMedCLIP** (medical-specific CLIP)
- **Healthy centroid baseline** computed from normal CXRs
- **Contrastive retrieval scoring** = visual similarity × abnormality deviation
- Fine-tuned **Phi-3-vision / LLaVA-Med** with LoRA for medical impression style
- Hallucination mitigation via retrieved evidence + LLM-as-a-judge

CONTRASTIVE RETRIVAL SCORING

After retrieving the top 10–15 similar cases by standard FAISS similarity, each case is re-ranked using a contrastive scoring formula that combines:

- Visual similarity between the query and the case
- Abnormality of the retrieved case (distance from healthy centroid)
- Abnormality of the query itself

The final score is:

$$\textbf{final_score} = \textbf{similarity} \times (1 + \lambda * \textbf{case_abn}) \times (1 + \mu * \textbf{query_abn})$$

Where:

similarity = cosine similarity

case_abn = abnormality score of retrieved case

query_abn = abnormality score of the input CXR

λ and μ are weights (e.g., 0.5 and 0.3)

OUTPUT

- Prioritizes cases that are both similar AND clinically abnormal
- Aligns abnormal cases with abnormal queries
- Filters out irrelevant “normal but similar-looking” images
- Produces medically meaningful retrievals for RAG

ARCHITECTURE

ASET

“Hospital archive”

- **Load MIMIC-CXR and IU X-Ray datasets.**
  - Extract report findings/impressions and clean text.
  - Filter out healthy cases using CheXpert labels.
  - Prepare metadata for each case.
- **Multimodal Embedding Generation**
  - Use BioMedCLIP to generate image embeddings and text embeddings.
  - Fuse them into a single multimodal vector for each case.
- **Healthy Baseline Computation**
  - Compute a healthy centroid from embeddings of healthy CXRs.
  - Precompute abnormality scores for all cases (distance from centroid).
- **RAG Store Construction**
  - Store all fused embeddings + metadata + abnormality scores in FAISS.

ARCHITECTURE

ASET

- **Input Handling & Embedding**
  - User uploads a CXR.
  - Preprocess and embed it using BioMedCLIP.
- **Query Abnormality Scoring**
  - Compute the query's abnormality score by comparing to the healthy centroid.
- **Contrastive Retrieval**
  - Retrieve top 10–15 similar cases from FAISS by similarity.
  - Re-rank them using:

    final_score = similarity × abnormality alignment
  - Select top 5 cases as evidence.
- **Multi-Label Classification**
  - Predict various diseases using a classification head on the query embedding.
- **LLM-Based Impression Generation**
  - Feed retrieved cases, abnormality scores, and predicted labels to the LLM.
  - LLM generates **Findings**, **Impression**, and a grounded explanation.
- **LLM-as-Judge (Hallucination Filter)**
  - A second LLM instance checks the factual correctness against retrieved evidence.
  - If factuality < threshold (e.g., 7/10):
    - regenerate impression
    - or flag inconsistencies.

RELATED WORKS

Paper / Method	Their Core Idea	How Our Work Is Different
CXR-RePaiR (2021)	Retrieval-based radiology report generation using a CLIP-style image–text model to fetch similar reports and generate final text.	Retrieves purely by **similarity**, not abnormality relevance. No **healthy baseline**, no **abnormality scoring**, no **contrastive re-ranking**, and limited multi-label prediction. We add *healthy centroid*, *contrastive scoring*, and *classification + RAG + LLM* pipeline.
X-TRA: Cross-Modal Retrieval Augmentation for CXR (2023)	Improves classification by retrieving similar cases using multimodal embeddings and adding retrieval features as input.	No **text generation**, no **LLM**, no **abnormality-aware retrieval**, and no **healthy baseline deviation**. They use retrieval to enhance classification; we use retrieval for **classification + explanation + report generation**.
CXR-CLIP / BioMedCLIP (2023–2024)	Large-scale vision–language pretraining for chest X-rays enabling strong image–text embeddings.	Embeddings only — no **RAG pipeline**, no **report generation**, no **contrastive retrieval**, and no **healthy baseline modeling**. We *build an entire diagnostic workflow* on top of embeddings.
LaB-RAG (2024)	Label-Boosted RAG for radiology report generation: retrieves cases using image embeddings + labels to guide LLM report generation.	Uses basic RAG; no **abnormality deviation** logic, no **healthy centroid**, no **contrastive scoring**, and no **LLM-as-Judge**. Our method grounds generation in **queries' abnormality**, improves factuality, and integrates **classification + retrieval + LLM refinement**.

PERT CHART

AMITY UNIVERSITY

ASET

PERT Chart/Schedule of project completion: -

S. No.	Milestones	Number of weeks														
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	Project Initiation and Planning															
2	Literature Review + Gap Analysis															
3	Dataset access and Setup															
4	Data Cleaning and Preprocessing															
5	Multimodal Embedding Pipeline Development															
6	Contrastive Fine-Tuning / Metric Learning + Centroid & Abnormality Scoring Framework															
7	Vector Indexing, FAISS Optimization, Retrieval Pipeline & Classifier Integration															
8	LLM Fine-Tuning + RAG Integration + Explanation Generation Pipeline															
9	Evaluation and Optimization															
10	Research paper, documentation, Plagiarism Check and Editing															

MINOR ACHIEVEMENTS

ASET

- A Multi-Stage Vision–Language Pipeline Was Proposed
  - Integrated ViT-B/32 for global feature extraction
  - YOLOv8 for pathology localization
  - DenseNet for regional feature refinement
  - Unified embeddings fused into a transformer decoder (RATCHET)
  - RAG-based retrieval from biomedical literature
  - BioGPT for impression generation

- The pipeline demonstrated a theoretical approach for combining multimodal visual encoding with medical knowledge retrieval, something uncommon in student research.
- Demonstrated the feasibility of modular AI systems, showing how each stage contributes to global → local → textual reasoning.
- Provided a groundwork for an interpretable explainable AI system, where bounding boxes, textual retrieval, and generated impressions work together.

MINOR LIMITAITONS

- **Model Limitations**
  - Pipeline too heavy for optimization (ViT + YOLO + DenseNet + RATCHET + BioGPT)
  - No end-to-end tuning possible due to resource constraints
  - Embedding fusion not fully validated
- **Retrieval Limitations**
  - RAG queries not domain-optimized
  - No fine-tuned retriever for radiology
  - Retrieval filtering + re-ranking not implemented
  - FAISS indexing not optimized for performance
- **LLM Limitations**
  - BioGPT not fine-tuned on target dataset
  - No contrastive grounding to reduce hallucination
  - No explanation or justification module

MAJOR OVERCOME

- **Better Architecture**
  - Switch to single multimodal embedding pipeline
  - Add contrastive similarity retrieval
  - Add centroid + abnormality scoring
- **Better Retrieval**
  - Use domain-trained embeddings
  - Proper FAISS index optimization
  - Structured, multi-step retrieval
- **Better LLM Usage**
  - LLM used for explanations, not full report generation
  - Add LLM-as-judge for retrieval validation
  - Add evidence-grounded reasoning

CONCLUSION

This major project transforms the insights from our minor review paper into a complete, reproducible, and novel AI system titled “AI-Powered Chest X-ray Multi-Label Classification and Impression Generation Using Medical Vision and Language Models”. By introducing contrastive healthy-guided multimodal RAG, we provide the first framework that explicitly combines similar-case retrieval with deviation-from-normal reasoning — directly addressing radiologist workflow needs while significantly improving clinical factuality and explainability.